

- Q.26 What is bleeding and segregation in concrete? How can these be minimized? (CO4)
- Q.27 Write two objectives of mix design. Name any three grades of concretes as per IS 456. (CO5)
- Q.28 What is bulking of sand? How is water absorption adjusted on site? (CO5)
- Q.29 Define admixtures and write two examples. (CO6)
- Q.30 What precautions are taken before, during and after cold weather concreting? (CO7)
- Q.31 Define fibre-reinforced concrete. Write any two advantages. (CO7)
- Q.32 What is silica fume concrete? Mention two of its uses and benefits. (CO7)
- Q.33 Write four precautions to be taken while storing cement at construction sites. (CO8)
- Q.34 What is batching of aggregate? Name two methods used. (CO8)
- Q.35 List any four equipment used to transport concrete. (CO8)

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain the properties of concrete in plastic and hardened states (CO4)
- Q.37 Write steps of nominal mix design as per IS 456. (CO6)
- Q.38 Describe operations involved in concrete handling from batching to compaction. (CO8)

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4th Sem / Civil, Brick Tech., Constr. Mgmt., Highway Engg

Subject:- Concrete Technology

Time : 3Hrs. M.M. : 100

SECTION-A

- Note:** Multiple choice questions. All questions are compulsory (10x1=10)
- Q.1 Concrete contains which of the following combination? (CO1)
- Cement, sand, and clay
 - Cement, sand, aggregates and water
 - Cement, water, and bricks.
 - Lime, water, and gypsum
- Q.2 Water-cement ratio directly influences. (CO4)
- Cost of concrete
 - Strength and workability
 - Colour of concrete
 - Drying Shrinkage
- Q.3 Cement used for faster setting in cold conditions is (CO2)
- White cement
 - Low heat cement
 - Quick setting cement
 - Portland pozzolana cement

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- Q.4 Initial setting time of OPC is (CO2)
 a) 10 minutes b) 60 minutes
 c) 30 minutes d) 600 minutes
- Q.5 Slump cone test gives information about (CO5)
 a) Durability b) Hardness
 c) Workability d) Curing
- Q.6 Rebound hammer test is (CO2)
 a) Non-destructive test b) Both (a) and (b)
 c) Destructive test d) None of the above
- Q.7 High water-cement ratio leads to (CO4)
 a) Reduced bleeding
 b) Decrease In workability
 c) Increase in strength
 d) Decrease in strength
- Q.8 IS 10262-2009 is related to (CO6)
 a) Concrete testing b) Admixtures
 c) Mix design d) Curing
- Q.9 Compacting factor test is used to find (CO5)
 a) Strength of cement
 b) Workability of concrete
 c) Consistency of cement
 d) Hardness of aggregates
- Q.10 Which one of the following is non-destructive test? (CO9)
 a) Flexural strength test
 b) Compression strength test
 c) Pulse velocity test
 d) None of the above

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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 PPC stands for _____. (CO2)
- Q.12 Name the code used for concrete mix design. (CO5)
- Q.13 Slump test is used to measure _____. (CO4)
- Q.14 Setting time of cement is measured using _____. (CO2)
- Q.15 Proper compaction increases concrete strength. (True/False) (CO8)
- Q.16 Super plasticizers are examples of _____. (CO6)
- Q.17 The defect in concrete due to water accumulation on surface is _____. (CO8)
- Q.18 Fly ash is added to concrete for _____. (CO7)
- Q.19 Water-cement ratio controls _____. (CO3)
- Q.20 RMC stands for _____. (CO7)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Define concrete and list its advantages (any five) (CO1)
- Q.22 What are the physical properties of cement? Explain any four. (CO2)
- Q.23 Classify aggregates according to size and shape. Explain with examples. (CO2)
- Q.24 What is grading of aggregates? How is fineness modulus used in grading? (CO2)
- Q.25 Define workability of concrete. List any four factors affecting it. (CO4)

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